The Gridded EIF Explorer

Data Validation Report

Environmental Inequality Lab

2026-09-01

Abstract

This report validates the pre-aggregated Gridded Environmental Impacts Frame data published by the Gridded EIF Explorer. It is generated by executing a registry of checks against the published product and rendering the results. Every figure and every table below is computed at build time; none is transcribed by hand. Checks that could not run are recorded as such and never counted as passing.

Generated	2026-09-01T12:31:37+00:00	Pipeline version	1.0.0
Commit	`6e8c6397d4ed`	Registry version	1.0.0
Contract version	1.1.0	Derived data version	v1
DuckDB	1.5.5	Python	3.12.14
Artifacts validated	the published product	Sections	A, C

Summary of results

No check failed. Counts by outcome: PASS 9, FAIL 0, WARN 1, INFO 3, SKIP 0.

ID	Result	Check	Tier
A1	INFO	Source file manifest	2
A2	WARN	Schema contract conformance	2
A3	PASS	Statistical invariants of the source data	2
A4	PASS	Published artifacts resolve and match their advertised size	2
A6	PASS	Version coherence across the four version lines	2
C1	PASS	Crosswalk coverage of every published year	2
C2	PASS	Each grid cell is assigned to exactly one unit	2
C3	INFO	Border-cell snapping, and how much population depends on it	2
C4	PASS	Hierarchical consistency between independently built crosswalks	2
C5	PASS	Grid geometry is as documented	2
C6	INFO	Drift in the populated cell set across years	2
C7	PASS	Fitness of the published crosswalk for its advertised purpose	2
C8	PASS	Spatial concentration of any dropped population	2

Tier indicates what a check reads: **0** — Local — registry and configuration only; **1** — Local artifacts — reads the build tree on disk, no network; **2** — Published product — reads the published catalog and the Census source files; **3** — External benchmarks — downloads independent published data.

The published product

This section is reference rather than verdict: it records what the product contains and what its files look like, so that the checks that follow can be read against a known inventory. Everything in it is read from the registry (`catalog/variables.yaml`) and the published catalog at run time.

Source

The Gridded Environmental Impacts Frame is an experimental Census Bureau product built from administrative records with privacy noise infused. Counts will not match the Decennial Census or the Population Estimates Program, and under- and over-coverage are both real. This report validates the pre-aggregated product built from it, not the source data itself.

Table 2: Source data

Property	Value
Citation	Voorheis, J., Colmer, J., Houghton, K., Lyubich, E., Munro, M., Scalera, C., and Withrow, J. (2025). "The Public-Use Gridded Environmental Impacts Frame."
File directory	https://www2.census.gov/ces/gridded_eif/
Landing page	https://www.census.gov/data/experimental-data-products/gridded-eif.html
Source data version	5.0
Experimental	yes
Grid resolution	0.01 degrees
Coordinate reference system	EPSG:4326
Centroid offset	0.005 degrees
Key columns	grid_lon, grid_lat
Key column type	string
Geographic coverage	CONUS, AK, HI

Datasets

Datasets not marked enabled are declared in the registry but not built. They are listed here because their presence is a deliberate decision — adding one should be a configuration change rather than a code change — and because a reader looking for pollution or housing data should be able to see that it is known about and not yet served. Preliminary years are built from partial administrative records, are subject to revision, and are never silently mixed into a range of final years.

Table 3: Datasets declared by the registry

Name	Label	Built	Final years	Prelim.	Dimensions
ageracesex	Population by age, race, and sex	yes	2000–2024	2025	race_ethnicity, age_group, sex
raceincome	Population by race and income decile	yes	2000–2024	2025	race_ethnicity, income_decile
pollutants	Air pollution exposure	no	1999–2023	—	—
extreme_weather	Extreme weather exposure	no	—	—	—
homevalue	Housing units by home value	no	2024–2024	—	—

Dimensions and category codes

The code is the literal string stored in the data and is what a query must match; the label is what the site displays. The two differ where a published code is misleading on its face — the age groups are stored as `Under 18`, `19-65` and `Over 65` but describe the bins under 18, 18–64 and 65 and over — and the code is never changed to match the label, because doing so would break every query written against the source files. Residual categories are retained as visible values and never folded into totals or dropped from denominators; a share computed from these tables must state its denominator.

Table 4: Race and ethnicity — codes as stored, labels as displayed

Code in the data	Label shown	Residual
White	White	
Black	Black	
Hispanic	Hispanic	
Asian	Asian	
AIAN	American Indian & Alaska Native (AIAN)	
Other/Unknown	Other or unknown	yes

Hispanic takes precedence; all other categories are non-Hispanic.

Table 5: Age group — codes as stored, labels as displayed

Code in the data	Label shown	Residual
Under 18	Under 18	
19-65	18–64	
Over 65	65 and over	
Missing Age	Age not reported	yes

Table 6: Sex — codes as stored, labels as displayed

Code in the data	Label shown	Residual
M	Male	
F	Female	
Missing Gender	Not reported	yes

Table 7: Household income decile — codes as stored, labels as displayed

Code in the data	Label shown	Residual
0	Decile 0	yes
1	1st (lowest)	
2	2nd	
3	3rd	
4	4th	
5	5th	
6	6th	
7	7th	
8	8th	
9	9th	
10	10th (highest)	

Equal-count national deciles of household AGI, recomputed each year. A decile count is a distributional profile, not an income measure.

Noise measures

Two columns ship with every population dataset. They are different estimators of the same quantity, not alternative encodings of it, and are never summed together or mixed within a single table. The raw measure is unbiased but can be negative at cell level; the post-processed measure pools small noisy cells by race within each one-degree grid point, redistributes them, and clamps residual negatives to zero, which makes it non-negative at the cost of moving mass toward categories that are sparse at cell level.

Table 8: Noise measures

Column	Label	Non-negative	Default	Recommended when
n_noise	Raw (noise-injected) counts	no	no	aggregate population above 6
n_noise_postprocessed	Post-processed counts	yes	yes	aggregate population below 6

The population threshold that selects the default measure is 600,000. Below it the post-processed measure has materially lowe above it the raw measure does.

Geography levels

A level marked partial does not tile the country: cells outside every unit are dropped rather than snapped to a nearest one, so totals at that level do not sum to the national total. A level marked as not built is declared but deliberately not served, and the registry records the reason as data rather than as a comment so the decision cannot drift out of the file unnoticed. Block group is absent by design: a 0.01 degree cell is about the size of a median block group, so that level would imply precision the privacy noise cannot support.

Table 9: Geography levels declared by the registry

Name	Label	Built	Units	Coverage	Map	Phase
nation	Nation	yes	1	complete	yes	1
state	State	yes	51	complete	yes	1
county	County	yes	3,144	complete	yes	1
cbsa	Metro / micropolitan area	yes	939	partial	yes	2
puma	PUMA	yes	2,400	complete	yes	2
czone2020	Commuting zone (2020)	yes	598	complete	yes	2
zcta	ZIP Code Tabulation Area	yes	33,000	partial	no	3
tract	Census tract	no	85,000	complete	no	3
cd	Congressional district	yes	435	complete	yes	4

Table 10: Geography caveats recorded in the registry

Level	Caveat
cbsa	Metro and micropolitan areas do not cover the entire country. Cells outside any CBSA are excluded, so CBSA totals do not sum to the national total.
zcta	ZCTAs approximate postal delivery routes, not neighborhoods or any administrative unit. High demand, weak conceptual fit.
tract	Tracts approach the size of a single grid cell. Measured at ZCTA (33k units) 1,143 units already rest on one cell; tracts are 2.6x finer, and dense urban tracts can be physically smaller than one 1.1 km2 cell, so centroid assignment either finds no cell or finds one whose residents mostly live elsewhere. Noise stops averaging out. Treat tract estimates as indicative, and prefer county or PUMA where either will serve.
cd	Districts are redrawn; the boundary vintage matters for comparisons.

Published files

One partition per dataset, geography and year, plus one all-years file per dataset and geography for time series — the latter exists because latency on a multi-year query is dominated by per-file round trips rather than by bytes. Both contain identical values. Crosswalks map each grid cell to a unit at every level and are published so that users can aggregate the Census source grids this site does not serve. Boundaries carry an identifier and geometry only, with no data values, so a new year of data never requires regenerating them. Every derived path is versioned, so a breaking change publishes alongside the old tree rather than invalidating a cited URL.

Table 11: Published partitions by dataset and geography

Dataset	Geography	Years	Files	Rows	Size (MB)
ageracesex	cbsa	2000–2025	26	970,409	10.500
ageracesex	county	2000–2025	26	2,987,203	29.400
ageracesex	czone2020	2000–2025	26	628,472	7
ageracesex	nation	2000–2025	26	1,778	0.100
ageracesex	puma	2000–2025	26	2,614,356	26.600
ageracesex	state	2000–2025	26	71,416	0.900
ageracesex	zcta	2000–2025	26	22,691,727	161.800
raceincome	cbsa	2000–2025	26	1,522,428	15.900
raceincome	county	2000–2025	26	4,699,270	42.700
raceincome	czone2020	2000–2025	26	949,048	10.200
raceincome	nation	2000–2025	26	1,710	0.100
raceincome	puma	2000–2025	26	4,086,670	38.900
raceincome	state	2000–2025	26	87,062	1.200
raceincome	zcta	2000–2025	26	34,405,579	220.800

Read from the published catalog, which is what the site serves. When the checks in this report were run against a local build rather than the published product, this inventory still describes what is published; the title page records which was validated.

Table 12: Other published artifacts

Kind	Files	Size (MB)
All-years files	14	578.700
Grid-cell crosswalks	7	—
Boundary files	6	—
Name lookups	7	—

Crosswalk, boundary and name file sizes are not recorded in the catalog and are omitted rather than estimated.

File schemas

Read from the files themselves rather than from documentation. A derived partition carries the geography identifier, one column per dimension of its dataset, both noise measures, and the number of grid cells the row was aggregated from — that last column is what lets a user judge whether a figure rests on enough cells for the noise to have averaged out. Coordinates in the crosswalk are stored as strings because the join against the source files is an equality test and float equality silently drops rows. The `snapped` flag marks cells whose centroid fell outside every unit and were assigned to the nearest one, so a user can see which assignments were inferred and decide for themselves.

Table 13: A derived partition — column layout

Column	Type
geo_id	VARCHAR
race_ethnicity	VARCHAR
age_group	VARCHAR
sex	VARCHAR
n_noise	DOUBLE
n_noise_postprocessed	DOUBLE
n_cells	BIGINT

Table 14: A published crosswalk — column layout

Column	Type
grid_lon	VARCHAR
grid_lat	VARCHAR
geo_id	VARCHAR
snapped	BOOLEAN

Aggregation rules

Cells are assigned whole to the geography containing their centroid, rather than apportioned by area. This is simple, reproducible, and matches what the source user guide's own examples do. Where a geography tiles the country, a cell whose centroid falls outside every unit is snapped to the nearest one: those cells are a systematic border artifact rather than scattered noise, and dropping them would undercount border communities specifically. Crosswalks are built from the union of populated cells across every published dataset-year, not from any single year, because the populated subset of the grid is not stable over time.

Table 15: Aggregation rules declared by the registry

Rule	Value
Assignment method	point_in_polygon_centroid
Crosswalk cell set	union_of_all_published_years
Snap unmatched cells to nearest	yes
Maximum snap distance	25,000 m (measured in EPSG:5070)
Residual categories visible	yes
Measures may be mixed in one table	no

A Provenance and integrity

A1 Source file manifest

Result: INFO

Every Census source file this product is built from is identified by URL, size, and publication date, so any figure in this report can be traced to the exact bytes it came from.

Method. One HTTP HEAD request per source file named by the registry, recording the reported size and modification date.

Quantity	Measured	Expected
Source files named by the registry	52	—
Reachable	52	52
Total size	3.550 GB	—

Table 16: Census source files consumed by the published product

Dataset	Year	Bytes	Last modified	File
ageracesex	2000	55,062,102	Tue, 11 Feb 2025 15:28:55 GMT	gridded_eif_pop_ageracesex_2000.parquet
ageracesex	2001	53,004,720	Tue, 11 Feb 2025 15:29:32 GMT	gridded_eif_pop_ageracesex_2001.parquet
ageracesex	2002	53,889,799	Tue, 11 Feb 2025 15:30:11 GMT	gridded_eif_pop_ageracesex_2002.parquet
ageracesex	2003	55,575,607	Tue, 11 Feb 2025 15:30:50 GMT	gridded_eif_pop_ageracesex_2003.parquet
ageracesex	2004	56,592,403	Tue, 11 Feb 2025 15:31:31 GMT	gridded_eif_pop_ageracesex_2004.parquet
ageracesex	2005	58,009,848	Tue, 11 Feb 2025 15:32:12 GMT	gridded_eif_pop_ageracesex_2005.parquet
ageracesex	2006	61,886,495	Tue, 11 Feb 2025 15:32:58 GMT	gridded_eif_pop_ageracesex_2006.parquet
ageracesex	2007	60,555,151	Tue, 11 Feb 2025 15:33:36 GMT	gridded_eif_pop_ageracesex_2007.parquet
ageracesex	2008	62,099,112	Tue, 11 Feb 2025 15:34:18 GMT	gridded_eif_pop_ageracesex_2008.parquet
ageracesex	2009	65,539,546	Tue, 11 Feb 2025 15:35:03 GMT	gridded_eif_pop_ageracesex_2009.parquet
ageracesex	2010	66,149,170	Tue, 11 Feb 2025 15:35:50 GMT	gridded_eif_pop_ageracesex_2010.parquet
ageracesex	2011	67,025,906	Tue, 11 Feb 2025 15:36:37 GMT	gridded_eif_pop_ageracesex_2011.parquet
ageracesex	2012	68,060,882	Tue, 11 Feb 2025 15:37:25 GMT	gridded_eif_pop_ageracesex_2012.parquet
ageracesex	2013	68,085,774	Tue, 11 Feb 2025 15:38:16 GMT	gridded_eif_pop_ageracesex_2013.parquet
ageracesex	2014	67,245,190	Tue, 11 Feb 2025 15:39:03 GMT	gridded_eif_pop_ageracesex_2014.parquet
ageracesex	2015	69,057,243	Tue, 11 Feb 2025 15:39:50 GMT	gridded_eif_pop_ageracesex_2015.parquet
ageracesex	2016	69,471,070	Tue, 11 Feb 2025 15:40:39 GMT	gridded_eif_pop_ageracesex_2016.parquet
ageracesex	2017	69,772,170	Tue, 11 Feb 2025 15:41:27 GMT	gridded_eif_pop_ageracesex_2017.parquet
ageracesex	2018	70,402,909	Tue, 11 Feb 2025 15:42:18 GMT	gridded_eif_pop_ageracesex_2018.parquet
ageracesex	2019	71,124,870	Tue, 11 Feb 2025 15:43:08 GMT	gridded_eif_pop_ageracesex_2019.parquet
ageracesex	2020	71,692,899	Tue, 11 Feb 2025 15:43:59 GMT	gridded_eif_pop_ageracesex_2020.parquet
ageracesex	2021	71,869,170	Tue, 11 Feb 2025 15:44:50 GMT	gridded_eif_pop_ageracesex_2021.parquet
ageracesex	2022	72,293,247	Tue, 11 Feb 2025 15:45:40 GMT	gridded_eif_pop_ageracesex_2022.parquet
ageracesex	2023	72,565,776	Tue, 11 Feb 2025 15:46:30 GMT	gridded_eif_pop_ageracesex_2023.parquet
ageracesex	2024	73,731,728	Tue, 22 Jul 2025 12:15:33 GMT	gridded_eif_pop_ageracesex_2024.parquet
ageracesex	2025	74,264,279	Tue, 22 Jul 2025 12:15:37 GMT	gridded_eif_pop_ageracesex_2025_realtime.parquet

continued

Dataset	Year	Bytes	Last modified	File
raceincome	2000	58,739,984	Tue, 11 Feb 2025 15:47:43 GMT	gridded_eif_pop_raceincome_2000.parquet
raceincome	2001	56,952,824	Tue, 11 Feb 2025 15:48:23 GMT	gridded_eif_pop_raceincome_2001.parquet
raceincome	2002	57,861,073	Tue, 11 Feb 2025 15:49:04 GMT	gridded_eif_pop_raceincome_2002.parquet
raceincome	2003	59,696,500	Tue, 11 Feb 2025 15:49:43 GMT	gridded_eif_pop_raceincome_2003.parquet
raceincome	2004	60,745,798	Tue, 11 Feb 2025 15:50:27 GMT	gridded_eif_pop_raceincome_2004.parquet
raceincome	2005	62,616,838	Tue, 11 Feb 2025 15:51:08 GMT	gridded_eif_pop_raceincome_2005.parquet
raceincome	2006	66,396,473	Tue, 11 Feb 2025 15:51:56 GMT	gridded_eif_pop_raceincome_2006.parquet
raceincome	2007	65,328,148	Tue, 11 Feb 2025 15:52:42 GMT	gridded_eif_pop_raceincome_2007.parquet
raceincome	2008	66,919,878	Tue, 11 Feb 2025 15:53:28 GMT	gridded_eif_pop_raceincome_2008.parquet
raceincome	2009	70,363,603	Tue, 11 Feb 2025 15:54:18 GMT	gridded_eif_pop_raceincome_2009.parquet
raceincome	2010	71,412,128	Tue, 11 Feb 2025 15:55:08 GMT	gridded_eif_pop_raceincome_2010.parquet
raceincome	2011	71,635,775	Tue, 11 Feb 2025 15:55:59 GMT	gridded_eif_pop_raceincome_2011.parquet
raceincome	2012	72,621,633	Tue, 11 Feb 2025 15:56:51 GMT	gridded_eif_pop_raceincome_2012.parquet
raceincome	2013	72,304,622	Tue, 11 Feb 2025 15:57:43 GMT	gridded_eif_pop_raceincome_2013.parquet
raceincome	2014	72,445,271	Tue, 11 Feb 2025 15:58:34 GMT	gridded_eif_pop_raceincome_2014.parquet
raceincome	2015	74,648,121	Tue, 11 Feb 2025 15:59:25 GMT	gridded_eif_pop_raceincome_2015.parquet
raceincome	2016	75,305,366	Tue, 11 Feb 2025 16:00:18 GMT	gridded_eif_pop_raceincome_2016.parquet
raceincome	2017	75,811,185	Tue, 11 Feb 2025 16:01:12 GMT	gridded_eif_pop_raceincome_2017.parquet
raceincome	2018	76,204,554	Tue, 11 Feb 2025 16:02:06 GMT	gridded_eif_pop_raceincome_2018.parquet
raceincome	2019	76,418,335	Tue, 11 Feb 2025 16:03:04 GMT	gridded_eif_pop_raceincome_2019.parquet
raceincome	2020	78,212,378	Tue, 11 Feb 2025 16:03:57 GMT	gridded_eif_pop_raceincome_2020.parquet
raceincome	2021	78,196,830	Tue, 11 Feb 2025 16:04:51 GMT	gridded_eif_pop_raceincome_2021.parquet
raceincome	2022	78,474,125	Tue, 11 Feb 2025 16:05:46 GMT	gridded_eif_pop_raceincome_2022.parquet
raceincome	2023	78,688,691	Tue, 11 Feb 2025 16:06:42 GMT	gridded_eif_pop_raceincome_2023.parquet
raceincome	2024	80,526,832	Tue, 22 Jul 2025 12:15:43 GMT	gridded_eif_pop_raceincome_2024.parquet
raceincome	2025	81,661,582	Tue, 22 Jul 2025 12:15:48 GMT	gridded_eif_pop_raceincome_2025_realtime.parquet

Interpretation. This check records rather than judges: it fails only if a file the registry names cannot be retrieved at all. A file that has been modified since a previous run will show a later modification date here, which is worth noticing — the Census Bureau revises these files — but is not itself an error. Content hashes are in A5; the Census file server publishes no checksum, so size and date are the strongest fingerprint available without downloading every file.

A2 Schema contract conformance

Result: WARN

> *Every source file matches the pinned schema contract — column names, column types, and category values — in every published year, not only the year the contract was derived from.*

Method. Each source file's schema is read and compared against `catalog/contracts/source-schema-v5.json`: required columns present and of the declared type, row count within the declared range, and no category value outside the declared set.

Quantity	Measured	Expected
Contract version	1.1.0	—
Describes source data version	5.0	—
Source files checked	52	—
Conforming	52	52
Violations	0	0
Warnings	1	—

Table 17: Schema contract conformance by source file

Dataset	Year	Rows	Result	Warnings
ageracesex	2000	15,437,397	pass	0
ageracesex	2001	14,920,696	pass	0
ageracesex	2002	15,194,761	pass	0
ageracesex	2003	15,668,722	pass	0
ageracesex	2004	16,013,889	pass	0
ageracesex	2005	16,422,385	pass	0
ageracesex	2006	17,425,242	pass	0
ageracesex	2007	17,160,284	pass	0
ageracesex	2008	17,623,304	pass	0
ageracesex	2009	18,533,927	pass	0
ageracesex	2010	18,701,284	pass	0
ageracesex	2011	18,919,413	pass	0
ageracesex	2012	19,178,189	pass	0
ageracesex	2013	19,214,376	pass	0
ageracesex	2014	19,076,041	pass	0
ageracesex	2015	19,454,193	pass	0
ageracesex	2016	19,650,827	pass	0
ageracesex	2017	19,884,417	pass	0
ageracesex	2018	20,059,598	pass	0
ageracesex	2019	20,196,044	pass	0
ageracesex	2020	20,553,710	pass	0
ageracesex	2021	20,607,480	pass	0
ageracesex	2022	20,780,507	pass	0
ageracesex	2023	20,849,590	pass	0
ageracesex	2024	21,135,424	pass	0
ageracesex	2025	21,225,308	pass	0
raceincome	2000	16,128,073	pass	0
raceincome	2001	15,763,720	pass	0
raceincome	2002	16,089,157	pass	0
raceincome	2003	16,631,104	pass	0
raceincome	2004	16,989,766	pass	0
raceincome	2005	17,445,348	pass	0

continued

Dataset	Year	Rows	Result	Warnings
raceincome	2006	18,389,387	pass	0
raceincome	2007	18,317,810	pass	0
raceincome	2008	18,768,687	pass	0
raceincome	2009	19,640,994	pass	0
raceincome	2010	19,841,032	pass	0
raceincome	2011	19,902,723	pass	0
raceincome	2012	20,272,858	pass	0
raceincome	2013	20,228,698	pass	0
raceincome	2014	20,431,741	pass	0
raceincome	2015	20,975,779	pass	0
raceincome	2016	21,177,785	pass	0
raceincome	2017	21,423,032	pass	0
raceincome	2018	21,619,442	pass	0
raceincome	2019	21,813,059	pass	0
raceincome	2020	22,206,164	pass	0
raceincome	2021	22,348,343	pass	0
raceincome	2022	22,467,718	pass	0
raceincome	2023	22,589,974	pass	0
raceincome	2024	23,066,667	pass	1
raceincome	2025	23,423,476	pass	0

Table 18: Contract warnings (non-fatal)

Dataset	Year	Detail
raceincome	2024	income_decile: contract categories absent from data [0]

Interpretation. A violation means the upstream schema has moved and the pipeline is building against an assumption that no longer holds. The correct response is to update the contract deliberately and bump its version, never to loosen the check until it passes. Warnings are a weaker signal: a category the contract declares but the data does not contain is usually a genuine absence in that year, and a column present in the data but absent from the contract may mean new data has become available.

A3 Statistical invariants of the source data

Result: PASS

> *In every source file the national total falls in a plausible band, the post-processed measure is nowhere negative, and post-processing preserves race totals — the three properties the contract asserts about the numbers themselves rather than the schema.*

Method. Each source file is aggregated to a national total for both noise measures and to race margins for each, then compared against the bands and tolerances declared in the schema contract. The bands are declared in the contract, not chosen here.

Quantity	Measured	Expected
Source files checked	52	—
Satisfying every invariant	52	52
Plausible national total, lower bound	150,000,000 people	—
Plausible national total, upper bound	360,000,000 people	—
Race-total preservation tolerance	1,000 people	—

Table 19: Statistical invariants by source file

Dataset	Year	National total (raw)	Result
ageracesex	2000	244,934,051	pass
ageracesex	2001	241,814,245	pass
ageracesex	2002	243,254,355	pass
ageracesex	2003	247,086,445	pass
ageracesex	2004	249,233,337	pass
ageracesex	2005	253,286,655	pass
ageracesex	2006	263,064,463	pass
ageracesex	2007	264,528,020	pass
ageracesex	2008	271,487,632	pass
ageracesex	2009	283,350,142	pass
ageracesex	2010	286,017,964	pass
ageracesex	2011	291,032,311	pass
ageracesex	2012	298,150,746	pass
ageracesex	2013	293,032,047	pass
ageracesex	2014	293,759,963	pass
ageracesex	2015	309,084,482	pass
ageracesex	2016	311,782,397	pass
ageracesex	2017	313,130,092	pass
ageracesex	2018	314,849,963	pass
ageracesex	2019	316,381,020	pass
ageracesex	2020	319,900,742	pass
ageracesex	2021	317,850,425	pass
ageracesex	2022	319,931,068	pass
ageracesex	2023	321,204,320	pass
ageracesex	2024	324,443,002	pass
ageracesex	2025	335,290,102	pass
raceincome	2000	244,944,587	pass
raceincome	2001	241,805,714	pass
raceincome	2002	243,244,348	pass
raceincome	2003	247,068,835	pass
raceincome	2004	249,250,588	pass
raceincome	2005	253,273,074	pass
raceincome	2006	263,062,318	pass

continued

Dataset	Year	National total (raw)	Result
raceincome	2007	264,513,354	pass
raceincome	2008	271,493,666	pass
raceincome	2009	283,351,932	pass
raceincome	2010	286,025,571	pass
raceincome	2011	291,021,174	pass
raceincome	2012	298,178,999	pass
raceincome	2013	293,035,401	pass
raceincome	2014	293,749,894	pass
raceincome	2015	309,092,093	pass
raceincome	2016	311,779,297	pass
raceincome	2017	313,139,027	pass
raceincome	2018	314,829,362	pass
raceincome	2019	316,380,089	pass
raceincome	2020	319,915,106	pass
raceincome	2021	317,831,591	pass
raceincome	2022	319,935,222	pass
raceincome	2023	321,219,128	pass
raceincome	2024	324,461,809	pass
raceincome	2025	335,278,878	pass

Interpretation. These catch a class of failure a column check cannot: the schema is intact but the numbers have moved. Post-processing redistributes population within race groups, so race margins must survive it; a departure means the upstream algorithm has changed and the guidance this site gives about choosing between the two measures needs revisiting.

A4 Published artifacts resolve and match their advertised size

Result: PASS

> *Every file the published catalog advertises exists at the URL given, returns HTTP 200, and is exactly the number of bytes the catalog claims.*

Method. One HTTP HEAD request per catalog entry — per-year partitions, all-years files, and crosswalks — comparing the reported content length against the size recorded in the catalog.

Quantity	Measured	Expected
Per-year partitions advertised	364	—
All-years files advertised	14	—
Crosswalks advertised	7	—
Resolving at the advertised size	385	385

Interpretation. The catalog is fetched by the site at runtime and is the only index of what exists. An entry that does not resolve is a broken download for a user; one whose size disagrees means the catalog and the object store have diverged, which usually indicates a publish that partially failed. A response carrying no content length is reported separately from a size mismatch, because that is a fact about the response rather than about the file.

A6 Version coherence across the four version lines

Result: PASS

> *The site, pipeline, schema contract, and derived data versions recorded in the published catalog agree with the repository they are supposed to have been built from, and every derived URL sits under the declared derived version prefix.*

Method. The four version fields in the published catalog are compared against the values the repository declares, and every entry URL is checked for the derived version prefix. The catalog committed to the repository is compared against the published one separately.

Quantity	**Measured**	**Expected**
Version lines disagreeing	0	0
Entry URLs outside the derived version prefix	0	0
Committed catalog differs from published	no	no

Table 20: Declared versions, repository against published catalog

Version line	**Repository**	**Published catalog**	**Agree**
pipeline version	1.0.0	1.0.0	yes
registry version	1.0.0	1.0.0	yes
derived version	v1	v1	yes
catalog version	1.1.0	1.1.0	yes

Interpretation. Disagreement means the published data was built from a different revision than the one being read, so nothing else in this report can be attributed to a known state of the code. The committed `site/catalog.json` is a development fallback only — the site fetches the live catalog at runtime — so a difference there affects local development rather than users, but anyone testing with `?catalog=./catalog.json` is validating against data that may no longer be published.

B Boundaries and geometry

No checks are implemented for this section yet. It is listed here so its absence is visible rather than silent.

C Crosswalk and cell assignment

C1 Crosswalk coverage of every published year

Result: PASS

The crosswalk assigns a geography to every populated grid cell that lies within the United States, in every published year. Aggregation is therefore lossless except for cells that fall outside the country by more than the declared snap radius, which are excluded deliberately and counted separately.

Method. Each source file is aggregated to one row per populated grid cell and joined against the county crosswalk. Cells with no match are collected across all dataset-years and adjudicated geometrically: the distance from each to the nearest county polygon is measured in EPSG:5070, and a cell counts as a legitimate exclusion only if that distance exceeds the snap radius the registry declares.

Quantity	Measured	Expected
Published dataset-years checked	52	—
Dataset-years with unassigned cells	4	—
Distinct cells unassigned	1	—
... beyond the snap radius, excluded deliberately	1	—
... inside the snap radius, unaccounted for	0	0
Snap radius	25 km	—
Farthest unaccounted-for cell from a polygon	n/a	—
Largest single-year shortfall	17 people	—
... as a share of that year's source total	0.000 %	—

Table 21: Grid cells and population unassigned by the crosswalk

Dataset	Year	Cells in source	Cells missing	People dropped	Loss (%)
ageracesex	2000	2,300,604	0	0	0
ageracesex	2001	2,180,995	0	0	0
ageracesex	2002	2,223,632	0	0	0
ageracesex	2003	2,286,972	0	0	0
ageracesex	2004	2,318,312	0	0	0
ageracesex	2005	2,366,497	0	0	0
ageracesex	2006	2,473,172	0	0	0
ageracesex	2007	2,430,241	0	0	0
ageracesex	2008	2,471,826	0	0	0
ageracesex	2009	2,558,298	0	0	0
ageracesex	2010	2,568,944	0	0	0
ageracesex	2011	2,551,866	0	0	0
ageracesex	2012	2,551,372	0	0	0
ageracesex	2013	2,547,992	0	0	0
ageracesex	2014	2,577,969	0	0	0
ageracesex	2015	2,600,809	0	0	0

continued

Dataset	Year	Cells in source	Cells missing	People dropped	Loss (%)
ageracesex	2016	2,602,674	0	0	0
ageracesex	2017	2,604,272	0	0	0
ageracesex	2018	2,607,807	0	0	0
ageracesex	2019	2,605,784	0	0	0
ageracesex	2020	2,612,800	0	0	0
ageracesex	2021	2,611,610	0	0	0
ageracesex	2022	2,611,734	0	0	0
ageracesex	2023	2,606,922	0	0	0
ageracesex	2024	2,606,191	1	17	0.000
ageracesex	2025	2,625,835	1	17	0.000
raceincome	2000	2,300,604	0	0	0
raceincome	2001	2,180,995	0	0	0
raceincome	2002	2,223,632	0	0	0
raceincome	2003	2,286,972	0	0	0
raceincome	2004	2,318,312	0	0	0
raceincome	2005	2,366,497	0	0	0
raceincome	2006	2,473,172	0	0	0
raceincome	2007	2,430,241	0	0	0
raceincome	2008	2,471,826	0	0	0
raceincome	2009	2,558,298	0	0	0
raceincome	2010	2,568,944	0	0	0
raceincome	2011	2,551,866	0	0	0
raceincome	2012	2,551,372	0	0	0
raceincome	2013	2,547,992	0	0	0
raceincome	2014	2,577,969	0	0	0
raceincome	2015	2,600,809	0	0	0
raceincome	2016	2,602,674	0	0	0
raceincome	2017	2,604,272	0	0	0
raceincome	2018	2,607,807	0	0	0
raceincome	2019	2,605,784	0	0	0
raceincome	2020	2,612,800	0	0	0
raceincome	2021	2,611,610	0	0	0
raceincome	2022	2,611,734	0	0	0
raceincome	2023	2,606,922	0	0	0
raceincome	2024	2,606,191	1	11	0.000
raceincome	2025	2,625,835	1	11	0.000

Loss is the share of that dataset-year's source national total that never reaches any published partition.

Interpretation. Aggregation inner-joins each year's source against the crosswalk, so a cell the crosswalk lacks is discarded silently along with its population — a partition built that way is a valid file with plausible numbers in it. Testing this on the year a crosswalk was built from cannot fail, which is why every published year is tested. Cells beyond the snap radius are outside the country and excluded on purpose; they are adjudicated by distance rather than against a list, so the check cannot be satisfied by a growing set of exceptions. Any unassigned cell inside the radius is population missing from the published totals.

C2 Each grid cell is assigned to exactly one unit

Result: PASS

> *No grid cell appears twice in a crosswalk. A duplicated cell would be counted once for each unit it appears under, inflating totals in a way that no total-preserving check would catch.*

Method. For each crosswalk, the number of rows is compared against the number of distinct grid cells.

Quantity	Measured	Expected
Crosswalks checked	7	—
Duplicate cell assignments	0	0

Table 22: Cell uniqueness by crosswalk

Geography	Rows	Distinct cells	Units	Duplicates
cbsa	2,021,331	2,021,331	925	0
county	3,021,989	3,021,989	3,144	0
czone2020	3,021,989	3,021,989	588	0
nation	3,021,990	3,021,990	1	0
puma	3,021,989	3,021,989	2,462	0
state	3,021,989	3,021,989	51	0
zcta	3,010,165	3,010,165	32,556	0

Interpretation. A cell whose centroid falls exactly on a shared polygon edge matches more than one unit in the spatial join; the pipeline resolves this by keeping the first match. This check confirms the resolution is applied consistently and that no duplicate survives into a published file. A duplicate would inflate the total of every geography it appears in while leaving the national total looking correct.

C3 Border-cell snapping, and how much population depends on it

Result: INFO

> *Cells whose centroid falls outside every polygon of a complete-coverage geography are recovered by snapping to the nearest unit rather than dropped, and the amount of population this recovers is stated rather than assumed.*

Method. The `snapped` flag carried by each crosswalk is counted per geography. The population riding on it is measured by joining the 2022 source file to the county crosswalk and summing the raw measure over snapped cells.

Quantity	Measured
Snapped cell assignments, all crosswalks	1,140
Population recovered at county level, 2022	14,214 people
Snap radius	25 km

Table 23: Snapped cell assignments by crosswalk

Geography	Coverage	Cells	Snapped	Snapped (%)
cbsa	partial	2,021,331	0	0
county	complete	3,021,989	285	0.009
czone2020	complete	3,021,989	285	0.009
nation	complete	3,021,990	0	0
puma	complete	3,021,989	285	0.009
state	complete	3,021,989	285	0.009
zcta	partial	3,010,165	0	0

Interpretation. Recorded rather than judged. Snapping applies only where a geography tiles the country completely, so an unmatched cell must belong somewhere; for partial-coverage geographies such as metropolitan areas an unmatched cell is a genuine absence and is dropped instead. The snapped cells are a systematic border artifact — cells straddling the Canadian and Mexican frontiers whose centroids land abroad — so dropping them would undercount border communities specifically, which for an environmental-inequality product are populations of interest.

C4 Hierarchical consistency between independently built crosswalks

Result: PASS

> *A cell contained by a county or PUMA polygon is contained by that unit's state: the first two digits of its identifier equal its state FIPS code. The crosswalks come from separate spatial joins against separate shapefiles, so agreement is an independent check on both.*

Method. Child and parent crosswalks are joined on grid cell and their identifiers compared. Disagreements are partitioned three ways: cells placed by containment in both crosswalks, cells where at least one placement came from nearest-polygon snapping, and cells where the source shapefiles themselves conflict — the last confirmed by re-running containment against the TIGER geometry rather than inferred.

Quantity	Measured	Expected
Nesting relationships checked	2	—
Disagreements explained by the TIGER layers	1	—
Disagreements involving a snapped placement	0	—
Unexplained disagreements between contained cells	0	0

Table 24: Nesting agreement between independently built crosswalks

Relationship	Cells compared	Contained, not nesting	Explained by TIGER	Snapped, not nest
county → state	3,021,989	0	0	
puma → state	3,021,989	1	1	

Interpretation. Containment is transitive and must nest: a cell inside a county polygon is inside that county's state as a matter of geometry, and a disagreement there means one of the two joins is wrong. Two other cases are not defects. Snapping is decided per level, so near a border the nearest PUMA and the nearest state may sit in different states without either result being incorrect. And TIGER's layers are published separately and are not guaranteed mutually consistent, most visibly along rivers whose channel has moved since the boundary was drawn; there both joins are correct with respect to their own input. Only the unexplained remainder counts against this check. Commuting zones are excluded entirely: they are derived by remapping county identifiers rather than by their own spatial join, so agreement with county is true by construction and would be evidence of nothing.

C5 Grid geometry is as documented

Result: PASS

Grid cell coordinates are centroids on a 0.01 degree lattice offset by 0.005, stored as strings, spanning the continental United States plus Alaska and Hawaii and no territory.

Method. Coordinate column types are read from the source schema. Every distinct cell in the 2022 file is tested arithmetically for the documented lattice offset, rather than by string suffix, so the test holds regardless of how many decimal places a file happens to print. Territory assignment is counted from the county crosswalk.

Quantity	Measured	Expected
Distinct populated cells, 2022	2,611,734	—
Grid resolution	0.010 degrees	—
Declared centroid offset	0.005 degrees	—
Departures from the documented specification	0	0

Table 25: Grid geometry against its documented specification

Property	Documented	Observed
Coordinate storage type	VARCHAR	grid_lon=VARCHAR, grid_lat=VARCHAR
Longitude off the 0.005 lattice	0 cells	0 cells
Latitude off the 0.005 lattice	0 cells	0 cells
Longitude range	CONUS + AK + HI	-176.655 to -66.955
Latitude range	CONUS + AK + HI	19.005 to 71.325
Cells assigned to a territory	0	0

Interpretation. Coordinates are stored as strings deliberately: the join between source and crosswalk is an equality test, and float equality silently drops rows. A departure from the lattice would mean the grid is not what the registry describes and the centroid-assignment method rests on a false premise. The longitude range spans the Aleutian crossing of the antimeridian, so its extremes are not a contiguous interval.

C6 Drift in the populated cell set across years

Result: INFO

The grid is fixed, but the populated subset of it is not: cells appear and disappear as administrative-records coverage changes and as people move. A crosswalk built from any single year therefore cannot assign another year's cells. This check measures how far the populated set drifts.

Method. For each published year of the age-race-sex dataset, the set of populated cells is compared against the 2022 file in both directions: cells that year has and 2022 does not, and cells 2022 has and that year does not. 2022 is a fixed point of comparison only; no year is privileged in the crosswalk itself.

Quantity	Measured
Years compared	26
Reference year for comparison	2,022
Most cells absent from the reference in one year	149,049 cells
Most cells present only in the reference	553,665 cells

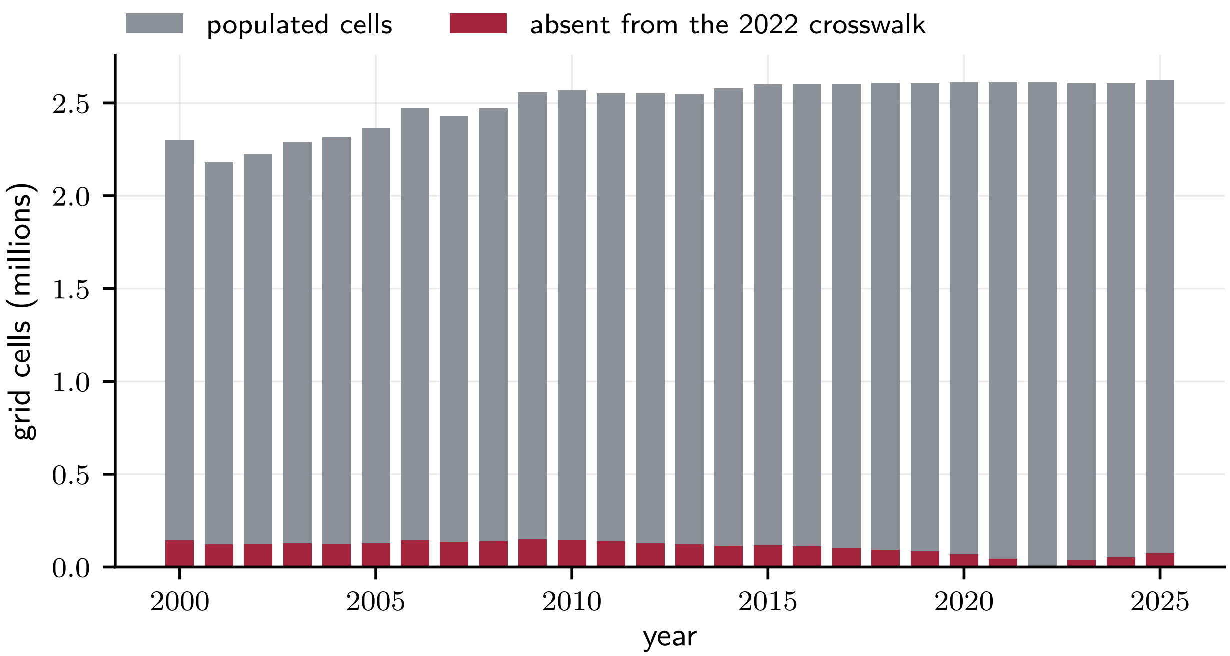

Figure 1: Populated grid cells by year. The dark band is the number of cells present in that year but absent from the 2022 file. A crosswalk built from the union of all years has no such band.

Table 26: Populated cell set by year, compared with the reference year

Year	Populated cells	Absent from 2022	Present in 2022 only
2000	2,300,604	143,487	454,617
2001	2,180,995	122,926	553,665
2002	2,223,632	124,894	512,996
2003	2,286,972	128,170	452,932

continued

Year	Populated cells	Absent from 2022	Present in 2022 only
2004	2,318,312	124,967	418,389
2005	2,366,497	128,389	373,626
2006	2,473,172	143,133	281,695
2007	2,430,241	134,996	316,489
2008	2,471,826	137,920	277,828
2009	2,558,298	149,049	202,485
2010	2,568,944	147,046	189,836
2011	2,551,866	137,895	197,763
2012	2,551,372	128,467	188,829
2013	2,547,992	122,300	186,042
2014	2,577,969	115,302	149,067
2015	2,600,809	116,307	127,232
2016	2,602,674	110,380	119,440
2017	2,604,272	102,829	110,291
2018	2,607,807	93,101	97,028
2019	2,605,784	84,717	90,667
2020	2,612,800	67,546	66,480
2021	2,611,610	44,941	45,065
2022	2,611,734	0	0
2023	2,606,922	39,364	44,176
2024	2,606,191	53,593	59,136
2025	2,625,835	75,125	61,024

A cell is populated in a given year if the source file contains any row for it. Drift runs in both directions, so no single year contains all the others.

Interpretation. Reported as a characterisation, not a pass or fail. Drift is a property of the source data — the Census Bureau is not doing anything wrong by publishing a moving cell set — and becomes a defect only if a pipeline assumes otherwise. Because the drift runs in both directions, no single year is a superset of the others, which is why crosswalks are built from the union of all published dataset-years. C1 and C7 are the checks that fail if that ever stops being true.

C7 Fitness of the published crosswalk for its advertised purpose

Result: PASS

> *The site offers the crosswalk so users can aggregate the Census source grids it does not serve. It should therefore cover the union of populated cells across every published dataset-year, not the cells of any one year.*

Method. The distinct cells of every published source file are unioned in a single scan and joined against the county crosswalk. Cells with no match are adjudicated by distance to the nearest county polygon, on the same basis as C1.

Quantity	Measured	Expected
Union of populated cells, all datasets and years	3,021,990 cells	—
Rows in the county crosswalk	3,021,989 cells	—
Union cells the crosswalk cannot assign	1 cells	—
. . . beyond the snap radius, outside the country	1 cells	—
. . . inside the snap radius, a genuine gap	0 cells	0

Interpretation. A user following the site's own instructions aggregates the raw grid with this file, so any cell it cannot assign is population that silently disappears from their analysis rather than ours. The pollutant and extreme-weather grids are not included in the union: they are not yet enabled in the registry and their cell coverage differs again, so a shortfall measured here is a lower bound on what a user of those files would encounter.

C8 Spatial concentration of any dropped population

Result: PASS

> *Population the crosswalk fails to assign is reported by state, so that a national figure cannot conceal a loss falling heavily on one place.*

Method. Cells unassigned in the most recent final year are located by point-in-polygon against the state boundaries and aggregated. The share is computed against what each state's total would have been had no cell been dropped, using the published state partition as the denominator.

Quantity	Measured	Expected
Year examined	2,024	—
Cells dropped	1	—
People dropped	17	—
Heaviest single dropped cell	17 people	—
Cells falling outside every state polygon	1	—
States affected	0	0

Interpretation. A national loss expressed as a fraction of a percent invites the reading that it is a rounding error, which holds only if the loss is diffuse. This check makes the distribution visible: a loss concentrated in particular states distorts the geographic pattern the product exists to describe whatever its national magnitude, and grid cells vary enormously in population, so a small number of dropped cells can carry a large number of people. Where every dropped cell falls outside all state polygons there is no distribution to report, and whether excluding those cells is correct is adjudicated in C1 rather than here.

D Numerical reconciliation

No checks are implemented for this section yet. It is listed here so its absence is visible rather than silent.

E External benchmarks

No checks are implemented for this section yet. It is listed here so its absence is visible rather than silent.

F Published claims and delivery

No checks are implemented for this section yet. It is listed here so its absence is visible rather than silent.

Reproducibility

This report is produced by `geif validate-report`. Each check is a registered function returning a typed result; the same registry drives the project's test suite, so a check cannot appear in this document without also being enforced in continuous integration.

Run environment

Repository commit	`6e8c6397d4ed62f8a1105d2ed4428743477870e7`
Working tree	clean
Platform	`linux`
Python	3.12.14
DuckDB	1.5.5
Tiers executed	0, 1, 2
Sections executed	A, C
Artifacts validated	the published product
Total check runtime	216.7 seconds

Queries

The queries behind the numbers above, with file locations replaced by placeholders. Source and published URLs are listed in the manifest tables of Section A.

C1 Crosswalk coverage of every published year

```
WITH src AS (
          SELECT grid_lon, grid_lat, sum(n_noise) AS pop
          FROM read_parquet('<source file>') GROUP BY 1, 2
       )
       SELECT count(*) AS cells,
             sum(src.pop) AS pop,
             count(*) FILTER (WHERE x.geo_id IS NULL) AS cells_missing,
             coalesce(sum(src.pop) FILTER (WHERE x.geo_id IS NULL), 0) AS pop_missing
       FROM src LEFT JOIN read_parquet('<crosswalk>') x USING (grid_lon, grid_lat)
```